

CE802 - QUESTIONS

This is a pdf version of a Moodle-based exam.

This exam is Open Book (Restricted), students may use the following in the exam: Formulary (available at the end of the exam).

Candidates are permitted to use: Calculator — Casio FX-83GT or Casio FX-85GT Series ONLY (CW, X, Plus)

Candidates must answer ALL QUESTIONS.

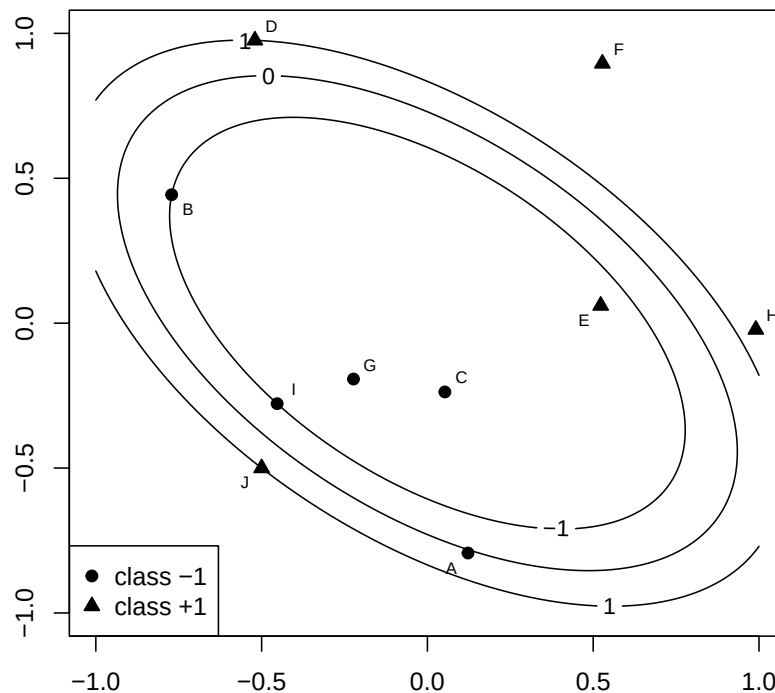
The percentages shown in brackets at the end of each question provide an indication of the proportion of the total marks for the PAPER which will be allocated.

Questions may be of type:

- **MCQ:** The student needs to select the only correct answer among a set of choices.
- **Cloze:** These are questions with gaps (highlighted in yellow) where the student is required to select an answer from a list of choices (the question shows the list of choices separated by “/”) or enter a numerical value (with the placeholder shown as “[?]”).
- **Open:** These are open questions where the student is asked to enter a response in a text box. Word limits are indicated in the question.

1. Question

The following plot shows the training instances, the decision boundary and the margin of a support vector classifier.



Which training instances are support vectors?

- The training instance A [?] is / is not? a support vector.
- The training instance B [?] is / is not? a support vector.
- The training instance C [?] is / is not? a support vector.
- The training instance D [?] is / is not? a support vector.
- The training instance E [?] is / is not? a support vector.
- The training instance F [?] is / is not? a support vector.
- The training instance G [?] is / is not? a support vector.
- The training instance H [?] is / is not? a support vector.
- The training instance I [?] is / is not? a support vector.
- The training instance J [?] is / is not? a support vector.

[10]

2. Question

Consider the following training instances, which consist of 2 features (x_1 and x_2) along with the corresponding ground truth r :

x_1	x_2	r
5	-3	-1
-6	-3	-1
2	7	-1
0	1	1
0	-6	1
-2	-6	1

- Now examine the linear separability of the training data (i.e. whether they can/cannot be classified with zero error using linear classifier). The points in the table above [?are / are not?] linearly separable. Hint: You find it helpful to plot the points on some scrap paper.
- Now consider the following nonlinear transformation: $\phi_1([x_1, x_2]) = [x_1 + 2|x_2|, x_2]$. Apply it to the original data in the table above and enter the missing values below:
 $\phi_1([5, -3]) = [11, -3]$.
 $\phi_1([-6, -3]) = [0, -3]$.
 $\phi_1([2, 7]) = [[?], [?]]$.
 $\phi_1([0, 1]) = [[?], [?]]$.
 $\phi_1([0, -6]) = [[?], [?]]$.
 $\phi_1([-2, -6]) = [[?], [?]]$.
 Now examine the linear separability of the transformed data. When transformed by the nonlinear mapping ϕ_1 the data points [?are / are not?] linearly separable.
- Now consider the following nonlinear transformation: $\phi_2([x_1, x_2]) = [x_1, x_2 + \frac{1}{2}x_1^2]$. Apply it to the original data in the table above and enter the missing values below:
 $\phi_2([5, -3]) = [5, 9.5]$.
 $\phi_2([-6, -3]) = [-6, 15]$.
 $\phi_2([2, 7]) = [[?], [?]]$.
 $\phi_2([0, 1]) = [[?], [?]]$.
 $\phi_2([0, -6]) = [[?], [?]]$.
 $\phi_2([-2, -6]) = [[?], [?]]$.
 Now examine the linear separability of the transformed data. When transformed by the nonlinear mapping ϕ_2 the data points [?are / are not?] linearly separable.

[17]

3. Question

A company wants to apply the agglomerative hierarchical clustering technique to cluster the objects in the table below, based on their attributes.

ObjectID	Weight	Material	Size	Pattern	Shape
1	light	wood	small	solid	cube
2	medium	wood	small	dotted	pyramid
3	heavy	metal	medium	checkered	sphere
4	light	wood	small	solid	sphere
5	medium	wood	medium	dotted	cube

The similarity of two objects is defined as the number of attributes with identical values. The similarity of two groups of objects is defined as the similarity of the most similar pair of objects where each member of the pair is drawn from a different group.

Construct a table showing the similarities (note: similarities, not distances) between each pair of objects among the 5 listed above.

Object ID	1	2	3	4	5
1	...	??	??	??	??
2	...	...	??	??	??
3	...	...	...	??	??
4	...	...	...	...	??
5	...	...	...	...	...

Afterwards, follow the agglomerative hierarchical clustering algorithm to cluster the data provided.

Initially each object is assigned to a separate cluster with the same ID as the object, e.g. object with ObjectID=1 is assigned to the cluster with ClusterID=1, etc. Then, the following merges are performed:

- At iteration 1 a cluster with ClusterID 6 is created by merging the clusters with ClusterID [3 and 5 / 1 and 3 / 2 and 5 / 2 and 3 / 3 and 4 / 1 and 4 / 2 and 4 / 1 and 2 / 4 and 5 / 1 and 5?], which have similarity 4.
- At iteration 2 a cluster with ClusterID 7 is created by merging the clusters with ClusterID [2 and 3 / 2 and 5 / 1 and 6 / 3 and 4 / 5 and 6 / 4 and 6 / 1 and 5 / 1 and 4 / 3 and 5 / 2 and 6?], which have similarity 3.
- At iteration 3 a cluster with ClusterID 8 is created by merging the clusters with ClusterID [4 and 5 / 3 and 6 / 1 and 7 / 1 and 3 / 5 and 7 / 3 and 5 / 3 and 4 / 2 and 3 / 1 and 2 / 6 and 7?], which have similarity 2.
- At iteration 4 a cluster with ClusterID 9 is created by merging the clusters with ClusterID [5 and 7 / 5 and 8 / 4 and 5 / 1 and 8 / 3 and 8 / 1 and 6 / 4 and 6 / 2 and 5 / 2 and 8 / 2 and 4?], which have similarity 1.

[18]

4. Question

You are a commuter relying on train travel from Town A to Town G and faced with two route options. Route one includes stops at intermediate towns B, C, and D, each segment requiring 1 hour ($A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow G$). Route two offers a quicker path from A to E, followed by a connection from E to F, then one from F to G, with each one of the three legs ($A \rightarrow E$, $E \rightarrow F$, $F \rightarrow G$) taking an hour each and minimal transfer time. However, town E hosts a poorly maintained train station lacking a reliable travel information system. When at town E, it is extremely hard to distinguish between boarding the correct train continuing to F or inadvertently boarding a train looping back to A (also taking one hour). The probability of boarding the correct train is represented by p .

You possess a monthly pass and are not required to pay for each train journey. Reaching town G is considered receiving a reward of 100, while the inconvenience of taking a longer route can later be accounted for by a discount factor.

Describe how this situation can be represented and modeled as a Markov Decision Process (MDP). Specifically, determine and calculate the elements of the tuple used to formally define an MDP: $\langle S, A, T(s'|s, a), R(s, a) \rangle$, including values of transition probabilities and rewards for each combination of their arguments (you can omit combinations that result in zero probability or zero rewards).

Note: Keep your response concise, ensuring it does not exceed 300 words. You can check the word count by clicking the “#” button on the editor toolbar.

[15]

5. Question

With regards to the previous question on the use of MDP to model the problem of a commuter relying on train travel from Town A to Town G, consider a scenario where the probability of boarding the correct train at town E (p) increases due to improvements in the travel information system. Discuss the implications of such an improvement.

In particular, consider the following points:

- How would an increase in p change the state transition probabilities?
- How would this, in turn, affect the state-action value functions for the choices faced by the commuter in Town A (i.e. $Q(A, take_route_1)$ and $Q(A, take_route_2)$)?
- Imagine p takes the value that makes $Q(A, take_route_1) = Q(A, take_route_2)$. Does this mean that route 1 and 2 are now completely equivalent? In particular, consider the uncertainty around the travel time and to what extent the state-action value function Q helps account for it.

Note: Keep your response concise, ensuring it does not exceed 250 words. You can check the word count by clicking the “#” button on the editor toolbar.

[10]

6. Question

In the “classical regime” of Machine Learning...

Bias² [?increases with / is independent of / decreases with / decreases and then increases with / increases and then decreases with?] the complexity of the model.

Variance [?increases with / is independent of / decreases with / decreases and then increases with / increases and then decreases with?] the complexity of the model.

The noise (or irreducible error) [?increases with / is independent of / decreases with / decreases and then increases with / increases and then decreases with?] the complexity of the model.

The prediction error on the training data [?increases with / is independent of / decreases with / decreases and then increases with / increases and then decreases with?] the complexity of the model.

The prediction error on test data [?increases with / is independent of / decreases with / decreases and then increases with / increases and then decreases with?] the complexity of the model.

[10]

7. Question

Training and Testing/Validation must use different data sets:

- (a) because estimating the accuracy using the same set used for training would very likely overestimate the true accuracy on future unseen data
- (b) because estimating the accuracy using the same set used for training would very likely underestimate the true accuracy on future unseen data
- (c) because using different datasets helps to increase the diversity of the data, which in turn improves the model's performance on the training data
- (d) because using different datasets helps to decrease the model's bias and make it more robust to changes in the underlying data distribution

[5]

8. Question

Consider the following Python code snippet used for machine learning:

```
from sklearn.model_selection import cross_val_score, GridSearchCV
from sklearn.svm import SVC
import pandas as pd

# Load dataset
df = pd.read_csv('data.csv')
# Create X (all columns except target)
X = df.drop('target', axis=1)
# Create y (target column)
y = df['target']

# Perform grid search within cross-validation for parameter tuning
model = SVC()
param_grid = {
    'C': [0.1, 1, 10, 100],
    'gamma': [0.001, 0.1, 1],
}
grid_search = GridSearchCV(model, param_grid, cv=5, scoring='accuracy')
grid_search.fit(X, y)

# Select best model
best_model = grid_search.best_estimator_

# Fit classifier using with the best parameters
# and assess its accuracy in cross-validation
scores = cross_val_score(best_model, X, y, cv=5, scoring='accuracy')

# Print average cross-validation accuracy
print('Cross-Validation_Accuracy:', scores.mean())
```

Carefully analyze the code and identify potential issues that may affect the reliability of the model training process.

Your answer should cover the following points:

- Identify the main potential issue that might affect the reliability of the model training process.
- Explain the implications of the identified issue.
- Propose strategies or approaches to address the identified issue and provide a detailed rationale for the proposed solution.

Note: Keep your response concise, ensuring it does not exceed 250 words. You can check the word count by clicking the “#” button on the editor toolbar.

[15]